D6. Audit Assessment Prompt Design

Prompt contract, alternative designs, selection rationale and safety handling.

Field	Value
Project	Development of AI-Based Cyber Security Audit Assessment Against Benchmark
Scope	Objective 3; secure local/air-gapped RAG proof of concept
Report date	28 August 2026
Evidence rule	Implemented, tested and untested claims are labelled separately.

Prompt comparison and final contract

Prompt	Core instruction	Accuracy	Output control	Decision
v1_direct	Directly answer from retrieved evidence	56.25%	Less explicit evidence review	Comparison baseline
v2_evidence_review	Answer, evidence IDs, support and justification	62.50%	Schema-constrained JSON	Selected

Why the final prompt is fit for audit scoring

The final response contract is intentionally restrictive: Yes, No or Insufficient Evidence; supporting evidence; valid retrieved chunk IDs; and a concise justification. That gives the scoring harness a stable output and prevents a fluent but untraceable paragraph from becoming an audit decision. [E8][E10]

v2 is selected both because it scored better on the controlled Qwen2.5 run and because it meets the assignment's mandatory evidence-based response requirement. Prompt injection in source text is treated as untrusted report content, while invalid citations are rejected by deterministic checks. [E7][E10]

Final prompt contract

The selected v2_evidence_review prompt defines an audit-assessment role, supplies a benchmark question and retrieved evidence, and requires a single structured result with answer, supporting evidence, evidence chunk IDs and justification. Answers are constrained to Yes, No or Insufficient Evidence. [E8]

The structured response also supports claim_evidence and abstention_reason fields for future verification without breaking the required output contract. [E8]

Prompt alternatives

Version	Design	Decision
v1_direct	Short direct assessment instruction	Retained as a comparison implementation [E8]
v2_evidence_review	Evidence-first JSON response; cites chunk IDs; requires justification	Selected implementation [E4][E8]
Future verifier prompt	Claim-by-claim evidence entailment and contradiction check	Optional implementation path; not calibrated on a large labelled set [E8]

Why v2 is selected

The task requires evidence-based answers and traceability to question-level audit decisions. v2 makes the evidence ID part of the machine-readable response, which enables deterministic citation-boundary validation. [E7][E8]

The two-report run used v2_evidence_review with schema enforcement, temperature 0.0, seed 42 and 384 output tokens. It completed without recorded execution errors. [E4]

Safety and limitation

Report content is evidence, not instruction. Prompt-injection-like text is detected and source boundaries are checked. Unknown citation IDs are rejected. [E7][E8]

The controlled Qwen2.5 prompt comparison gives v2 62.50% accuracy and v1 56.25%. v2 is retained because it improves the observed control result and preserves the structured evidence contract. [E10]

Evidence

The exact prompt implementation is in src/prompts.py and the use of v2 is captured in run_config.json. [E4][E8]

Final response schema

Field	Purpose	Validation rule
answer	Benchmark decision	Only Yes, No or Insufficient Evidence.
supporting_evidence	Short report-grounded quote/paraphrase	Empty when evidence is insufficient.
evidence_chunk_ids	Links decision to returned context	Every ID must belong to retrieved evidence.
justification	Explains evidence-to-verdict link	One or two concise sentences.
abstention_reason	Makes uncertainty explicit	Required for an Insufficient Evidence outcome.

Prompt-selection judgement

The v2 prompt is not simply longer wording. It adds an explicit evidence-review instruction before the schema response, asks the model to cite only checked chunks, and separates abstention from unsupported guessing. Those instructions are aligned with the assignment's requirement to preserve evidence behind a benchmark decision. [E8]

Under the Qwen2.5 control, v2 scored 62.50% accuracy versus 56.25% for v1. It is therefore selected because it improves the observed score and enables deterministic validation; it is not claimed to solve factual entailment, especially where retrieved context is incomplete. [E7][E10]

Engineering rationale and review use

Prompt design is not a cosmetic part of this system. The prompt is the control plane between retrieved audit evidence and the audit-level score, so it must force the model to expose the answer category, quoted support and source identifiers in a form the harness can validate. A prose-only answer would be easier to demo but would break traceability and make error analysis much weaker. [E5][E8]

The prompt also makes missing evidence a first-class result rather than an error condition. This matters for audit assessment because an unsupported positive claim can falsely inflate the final score. The present dataset cannot yet measure abstention quality because it has no Insufficient Evidence gold examples; the implementation already reserves the category and the next benchmark must evaluate it directly. [E5][E8]

For operational use, prompt versions should remain frozen during every comparative experiment. Any change to instructions, schema, demonstration count or evidence format must be logged alongside model, retrieval, seed and question-set changes. The current harness design supports that discipline through stored configuration and question-level artifacts. [E4][E8]

Repository evidence cited

- [E4] results/production_real_reports/1787846335_de0ded86a7db/run_config.json and run.log: two-report production-parser run configuration and execution log.
- [E7] Verification executed 28 August 2026: python -m unittest test_json_parser.py test_adaptive_rag.py test_security_gates.py (6 tests passed) and python test_harness_mock.py (offline mock end-to-end passed).
- [E8] src/{docling_parser.py,chunking.py,embed_index.py,pipeline.py,evaluation_harness.py,adaptive_rag.py,prompts.py,icl.py}: implementation inspection.
- [E10] results/final_validation/VALIDATION_SUMMARY.md and its cited run directories: controlled local LLM, ICL, prompt, retrieval, CRAG, chunking and chunk-parameter comparisons completed 28 August 2026.